Assignment – Part B: Individual Reflection Presentation Transcript

Title: Artificial Intelligence (AI) in the Enterprise: Innovation, Risk, and Responsible Governance

01 Introduction: From Speculation to Reality

Artificial intelligence (AI) has moved beyond speculation in many enterprise contexts and is becoming a strategic capability. As Kneusel (2023, Chapter 3) notes, with large datasets, “awe-inspiring things can happen.” Organisations now use AI to improve efficiency through automation, predictive analytics, and personalisation. Stanford HAI (2026, p. 193) reports that AI use among surveyed organisations rose from approximately 20% in 2017 to 88% in 2025. However, this rapid adoption has created a governance gap as regulation, assurance, and organisational controls are struggling to keep pace (Taeihagh, 2021). Therefore, the key question is how enterprises can capture AI’s benefits while managing ethical, security, and accountability risks.

02 Technical Overview: What is AI?

Russell and Norvig (2022, p. 19) describe AI as a universal field relevant across many domains of human thought and activity. At its core, AI involves systems that support reasoning, learning, decision-making, or action. Machine Learning (ML) identifies patterns in data and supports enterprise uses such as credit scoring, fraud detection, and churn prediction. Deep Learning extends this through neural networks enabling image recognition, speech processing, and natural language processing. Large Language Models (LLMs), such as ChatGPT and Claude, generate, summarise, and

analyse text at scale, giving them Generative AI (Gen-AI) capabilities. Agentic AI goes further by connecting models to tools and workflows. As autonomy increases, enterprises require stronger governance, monitoring, and accountability.

03 Enterprise Applications of AI

AI now spans multiple industries and business functions supporting automation, decision-making, and personalisation at scale. I will focus on three applications that demonstrate both business value and governance risk. A more comprehensive list is provided in Appendix 4. In financial services, AI supports fraud detection, credit risk assessment, and forecasting. This enables faster decisions but raises transparency and accountability concerns. In Human Resources, CV screening promises efficiency but can reproduce discrimination where models learn from biased historical data. Dastin's (2018) reporting on Amazon's recruitment algorithm and Mehrabi *et al.*'s (2021) survey of algorithmic bias illustrate this risk. Knowledge Management tools improve summarisation and retrieval, but introduce hallucination risks as shown later in the Deloitte Australia case (see Slide 07). Every application therefore represents not only a business opportunity but also a governance responsibility.

04 Ethical Issues

AI within organisations creates ethical challenges around fairness, accountability, explainability, and privacy. Leslie (2019) argues that responsible AI requires ethical and safety principles to be embedded into system design. Using financial services as an example, this slide shows where enterprise AI deployment can fall short of these

principles. Algorithmic bias undermines fairness because models trained on historical data can reproduce and amplify discrimination (Mehrabi *et al.*, 2021). Proxy discrimination challenges accountability as neutral variables such as postcode, device type, and transaction behaviour can encode protected characteristics indirectly (Langenbucher and Corcoran, 2022). Opacity creates explainability concerns because credit-scoring models may deliver faster decisions while reducing transparency and auditability (Bücker *et al.*, 2022). This raises issues under GDPR and FCA Consumer Duty expectations around fair and understandable outcomes. Finally, privacy risks challenge autonomy where personal data is used without transparency (Floridi *et al.*, 2018). Together, these failures represent not only a technology problem but also a governance problem—one that the frameworks outlined in Slide 06 are beginning to address.

05 Security Concerns

Adversarial attacks are a distinctive AI security threat. Research shows that deliberately manipulated inputs can cause models to misclassify while maintaining high confidence (Goodfellow, Shlens, and Szegedy, 2015). In financial services, this could allow attackers to bypass fraud detection or credit-scoring systems. Data leakage through inference attacks can expose sensitive training data, creating privacy and regulatory risks. Hallucination is especially relevant to this presentation (see Slide 07). In Natural Language Processing, a hallucination refers to content generated that may appear plausible but is inaccurate or unsupported by source content (Ji *et al.*, 2023, p. 4). Model poisoning creates further risk by corrupting training data and embedding errors

before deployment. These risks do not make AI unusable, but they do show why enterprises need adversarial testing, access controls, monitoring, and human review.

06 Regulation and Governance

This slide brings together the ethical and security risks discussed so far and considers how they can be governed. No single framework provides complete AI governance, so enterprises must combine legal, regulatory, and assurance-based controls. The EU AI Act (2024) classifies certain AI systems as high-risk and requires stronger controls (Regulation (EU) 2024/1689, 2024, Chapter III). However, its EU focus and phased implementation mean that organisations still require internal governance. GDPR and UK GDPR safeguards also require lawful processing, transparency, human review, and routes to challenge significant automated decisions. In financial services, the FCA Consumer Duty (2023) adds a customer-outcomes lens requiring fair, understandable, and appropriate decisions supported by testing, validation, and explainability (Financial Conduct Authority, 2024, p. 4). Finally, UK AI Assurance Guidance provides a practical bridge between principles and deployment (Department for Science, Innovation and Technology, 2023). In practice, this means using assurance techniques such as impact assessments, bias testing, red-teaming, audits, monitoring, and human review. These techniques reduce rather than eliminate risk, and their effectiveness depends on how rigorously they are implemented. In summary, governance should be risk-based, auditable, and embedded before AI systems are deployed, rather than added only after harm occurs.

07 Case Study: Deloitte Australia Gen-AI “Hallucination”

The risks discussed in the previous slides are not merely theoretical. In 2025, Deloitte Australia was required to correct and republish a report worth $290,000 after an independent researcher identified hallucinations, including fabricated citations and a false legal quotation (Kissin, 2025; Paoli, 2025). The consequences were significant: the Big Four firm partially refunded its client, publicly acknowledged the error, and suffered reputational damage. The implications for practice are clear. Organisations must verify AI-generated outputs before publication, maintain robust audit trails and source traceability, disclose Gen-AI use where appropriate (Deloitte Australia, 2025, p. 147), and retain human accountability for final assurance. This case study demonstrates that even large professional firms remain exposed when governance, oversight, and assurance controls are inadequate.

08 Generative AI (Gen-AI) Response Summary

To evaluate Gen-AI in professional practice, I submitted the assignment question to Perplexity AI. The full prompt, generated response, and cited sources are provided in Appendices 1 to 3. I selected Perplexity because it produces structured answers with visible source links, allowing both its content and supporting evidence to be assessed. The response was generated rapidly and followed a four-part structure: an introduction, critical evaluation, enterprise balance, and a final bottom line. It identified enterprise benefits including productivity, decision-making, and threat detection, alongside the broader trade-off between innovation and governance. It also discussed ethical risks such as bias, privacy, and transparency, as well as security threats including phishing,

impersonation, data leakage, and model poisoning. Its proposed controls included secure-by-design development, monitoring, and human oversight. These themes provided a useful initial overview which are critically evaluated in the next slide.

09 Critical Comparison: Human Research vs. Perplexity (Gen-AI) Response

The Perplexity response identified relevant themes quickly but lacked the academic depth and regulatory nuance required for professional judgement. On source reliability, it relied on mixed web and industry sources, which alone would not provide sufficient evidence for rigorous academic or regulated professional use. My research instead drew on academic, regulatory, and case-study evidence. The response also provided a broad summary rather than a sustained critical evaluation. Its uniform four-line paragraph structure appears template-driven, while its fluent but largely unsupported claims illustrate the wider concern identified by Bender *et al.* (2021) that large language models can generate plausible text without grounded understanding or domain reasoning. Ethical and regulatory gaps were also significant. No named ethical framework was applied, whereas my research used Leslie's (2019) responsible AI principles to link theory to practice. Similarly, the response did not engage with the EU AI Act, GDPR, FCA Consumer Duty, or UK AI Assurance Guidance. Finally, it included no detailed case study, while my research used the Deloitte Australia hallucination case to demonstrate how generative AI risks can affect professional practice. Perplexity was substantially faster and generated a useful initial overview, particularly around cybersecurity. My human-led research was slower and remained vulnerable to source-

selection bias and the limits of my own knowledge. However, it allowed greater regulatory precision, source verification, and contextual judgement.

10 Conclusions and Recommendations

To conclude, AI offers substantial enterprise value through automation, improved decision-making, and greater productivity. However, these benefits carry significant risks. These risks—bias, opacity, privacy loss, and weak accountability—can be structural features of AI rather than incidental flaws. The Deloitte Australia case demonstrates that an AI-related error can also become a failure of professional governance when verification, traceability, and accountability are inadequate. Enterprises should therefore treat responsible AI as a design requirement rather than an afterthought. High-impact systems require proportionate, risk-based assurance, including bias testing, privacy review, red-teaming, audit trails, continuous monitoring, and meaningful human oversight. Gen-AI can support research and initial scoping, but final professional judgement must remain human-led. Responsible governance can therefore support innovation; it enables AI to be deployed in a more trustworthy, auditable, and sustainable way.

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